



Lab 01: Basic LAN Setup

2911 Router · 2x 2960 Switches · PC0 & PC1

Lab #01

Topic: Networking
Fundamentals

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Build a LAN using two Cisco 2960 switches, a 2911 router and two PCs.
- Configure static IP addresses on both PCs with correct default gateways.
- Bring up router GigabitEthernet interfaces and verify Layer 3 connectivity.
- Use **ping** to confirm PC0 and PC1 communicate through the router.
- Relate to Network+ objective 2.2: Compare and contrast networking devices.

2. Network Topology

Router0 (2911) sits at the top. Gig0/0 connects to Switch0 (2960-24TT) which serves PC0. Gig0/1 connects to Switch1 (2960-24TT) which serves PC1. Two separate /24 subnets exist — one per router interface.

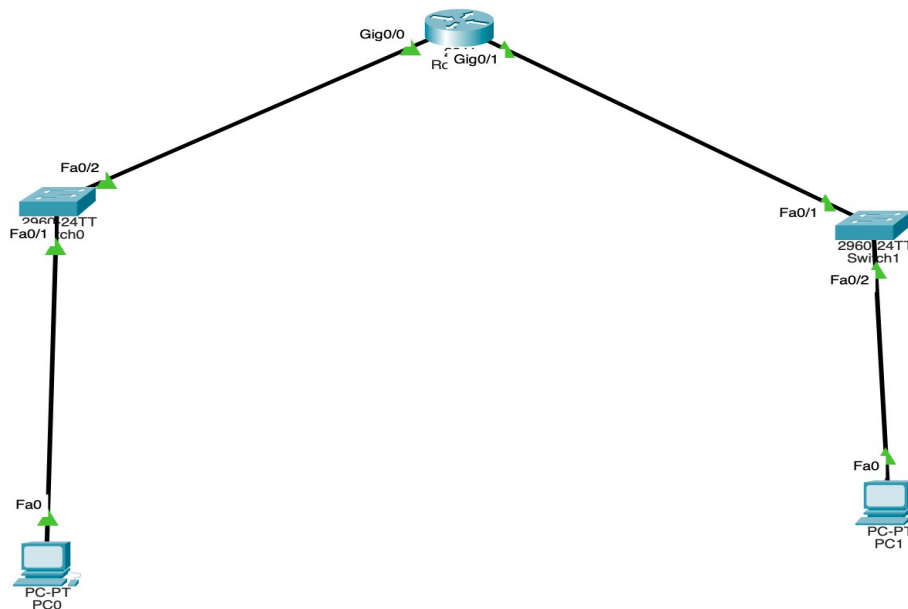


Figure 1 – Router0 (2911) with Switch0 & Switch1 (2960), PC0 and PC1

Device	Model	Interface	IP Address	Subnet Mask	Default GW
Router0	2911	Gig0/0	192.168.1.1	255.255.255.0	—
Router0	2911	Gig0/1	192.168.2.1	255.255.255.0	—



Switch0	2960-24TT	—	—	—	—
Switch1	2960-24TT	—	—	—	—
PC0	PC-PT	Fa0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	PC-PT	Fa0	192.168.2.10	255.255.255.0	192.168.2.1

3. Configuration

3.1 Router0 Interfaces

```
Router>enable
Router#configure terminal
Router(config)#hostname Router0
Router0(config)#interface GigabitEthernet0/0
Router0(config-if)#ip address 192.168.1.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#interface GigabitEthernet0/1
Router0(config-if)#ip address 192.168.2.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0#write memory
```

3.2 PC Static IPs

PC0 IP	192.168.1.10
PC0 Mask	255.255.255.0
PC0 GW	192.168.1.1
PC1 IP	192.168.2.10
PC1 Mask	255.255.255.0
PC1 GW	192.168.2.1

4. Verification

```
PC0>ping 192.168.2.10
PC0>ping 192.168.1.1
Router0#show ip interface brief
Router0#show ip route
```

5. Key Takeaways



- The 2911 uses GigabitEthernet — faster than the older 1841 FastEthernet.
- Each router interface = one subnet; two interfaces = two broadcast domains.
- Switches need no IP for Layer 2 forwarding — only for management (SVI).
- Interfaces default to 'shutdown' — always use **no shutdown** to activate.
- Verify with **show ip interface brief**: both interfaces must show up/up.